Series

Questions 1–123

1.09 — Question 1. Find:

a) $\sum_{r=1}^{30} 8$

b) $\sum_{r=1}^{50} r$

c) $\sum_{r=1}^{20} r^2$

d) $\sum_{r=1}^{10} r^3$

1.09 — Question 2. Find an expression for each of the following using standard series formulae. Fully expand your answer:

a) $\sum_{r=1}^{n} (2r + 3)$

b) $\sum_{r=1}^{n} (5r - 12)$

c) $\sum_{r=1}^{n} (4r + 1)$

d) $\sum_{r=1}^{n} (ar + b)$
where a and b are constants.

1.09 — Question 3. Find the positive integer k such that

$$\sum_{r=1}^{k}(2r+k)=300$$

1.09 — Question 4. Using the standard results for summations, show that

$$\sum_{r=1}^{12}r(6r-1)=3822$$

1.09 — Question 5. Using the standard results for summations, find the value of

$$\sum_{r=1}^{50}(3r+2)^2$$

1.09 — Question 6. Fully factorise each of the following:

a) $n(n+1)+4n$
b) $\frac{7}{2}n(n+1)-5n$
c) $3n(n+1)(2n+1)+\frac{1}{2}n(n+1)$
d) $4n(n+1)(2n+1)-\frac{1}{4}n^2(n+1)^2$
e) $\frac{1}{2}n^2(n+1)^2-3n(n+1)$
f) $\frac{1}{4}n^2(n+1)^2-\frac{1}{6}n(n+1)(2n+1)$
g) $\frac{3}{4}n^2(n+1)^2+\frac{5}{2}n(n+1)$

1.09 — Question 7. Find a fully factorised expression for each of the following using standard series formulae:

a) $\sum_{r=1}^{n}(r^3+r^2)$

b) $\sum_{r=1}^{n}(2r^2+5r)$

c) $\sum_{r=1}^{n}(7r^3-2r^2)$

d) $\sum_{r=1}^{n}(1-2r+5r^3)$

e) $\sum_{r=1}^{n} r(2r+1)$

f) $\sum_{r=1}^{n} r^2(r-4)$

1.09 — Question 8. Using the standard results for summations, show that

$$\sum_{r=1}^{n}(4r-2)^2 = \frac{4}{3}n(2n-1)(2n+1)$$

1.09 — Question 9. Using the standard results for summations, find the sum of the first n terms of each of the following series:

a) $(1 \times 2) + (2 \times 3) + (3 \times 4) + \ldots$
b) $(3 \times 5) + (4 \times 6) + (5 \times 7) + \ldots$
c) $(2 \times 4) + (3 \times 5) + (4 \times 6) + \ldots$
d) $2 \times 1^2 + 3 \times 2^2 + 4 \times 3^2 + \ldots$
e) $(1 \times 3 \times 4) + (2 \times 4 \times 5) + (3 \times 5 \times 6) + \ldots$

1.09 — Question 10.

a) Using the standard results for summations, find an expression for

$$\sum_{r=1}^{n} (r+1)(3r-1)$$

b) Hence evaluate

$$2 \times 2 + 3 \times 5 + 4 \times 8 + \ldots + 51 \times 149$$

1.09 — Question 11. Find a fully factorised expression for

a) $\sum_{r=1}^{2n} r^2$

b) $\sum_{r=1}^{2n} (2r)^2$

c) $1^2 + 3^2 + 5^2 + \ldots + (4n-1)^2$

1.09 — Question 12. Using the standard results for summations, show that

a) the sum of the square of the first 10 consecutive even numbers is 1540

b) the sum of the square of the first 10 consecutive odd numbers is 1330

c) the sum of the cubes of the first 10 consecutive even numbers is 24 200

1.09 — Question 13. Find the sum of all the integers from 1 to 999 inclusive that are not square numbers.

1.09 — Question 14. Given that the sum of the squares of the first n consecutive even numbers is 82 160, find the value of n

1.09 — Question 15. Given that the sum of the squares of the first n consecutive odd numbers is 20 825, find the value of n

1.09 — Question 16.

a) By considering the standard results for summations, find

$$24 + 3\sum_{r=1}^{n}(r+2)(r+3)$$

as a product of three linear factors.

b) Hence, show that

$$24 + 3\sum_{r=1}^{n}(r+2)(r+3)$$

is divisible by 6

1.09 — Question 17.

a) Show that

$$\sum_{r=1}^{n}(7r-2)^2 = \frac{1}{6}n(an^2+bn+c)$$

where a, b and c are constants to be found.

b) Hence find the value of k for which

$$\sum_{r=1}^{k}(7r-2)^2 = 190k^2$$

1.09 — Question 18. Given that $\mathrm{f}(r) = ar + b$ for constants a and b,

$$\sum_{r=1}^{3}\mathrm{f}(r) = 0$$

and

$$\sum_{r=1}^{7}\mathrm{f}(r) = 35$$

a) Find a general expression for

$$\sum_{r=1}^{n}\mathrm{f}(r)$$

b) Hence find

$$\sum_{r=1}^{20}\mathrm{f}(r)$$

1.09 — Question 19.

a) Show that

$$\sum_{r=1}^{n}(5r-7)=\frac{1}{2}n(5n-9)$$

b) Find the smallest value of k for which

$$\sum_{r=1}^{k}(5r-7)>6000$$

1.09 — Question 20. Find constants a and b such that

$$\sum_{r=1}^{n}(ar+b)=n^2-10n$$

1.09 — Question 21. An art gallery has a collection of n paintings numbered from 1 to n, which are rectangular with lengths $(r + 10)$ cm and $(r + 12)$ cm, where r is the number of each painting.

a) Show that the formula for the total area of n paintings is $\frac{1}{6}n(2n^2 + 69n + 787)$ cm^2.

b) Given that the total area of the n paintings is 3806 cm^2, calculate how many paintings there are in the collection.

1.09 — Question 22. A company manufacturing bicycles agrees to a 15-year contract with a supplier to provide bicycles annually. In the first year of the contract, 250 bicycles are delivered and each subsequent year the delivery increases by 40.

a) State an expression for the number of bicycles supplied in year r

In year r the price per bicycle (£) is given by $50 + \frac{r^2}{30}$ and the total cost of producing all the bicycles delivered that year (in £) is given by

$$2000 + 7r^2$$

b) Show that the profit made by the company in year r is

$$\frac{4}{3}(r^3 + 1500r + 6375)$$

c) Using the standard results for summations, show that the total profit made by the company in the first n years is

$$\frac{1}{3}n(n^3 + an^2 + bn + c)$$

where a, b and c are constants to be found.

After 15 years the actual profit of the company is £280 000

d) Comment on the validity of the model.

Prerequisite knowledge for Q23: Standard Deviation

1.09 — Question 23. A bakery is preparing a special display of n loaves of bread for the shop window. The loaves are arranged in ascending order of weight. The first loaf weighs 400 grams and each subsequent loaf is 3 grams heavier.

a) State the mass of the rth loaf in terms of r

b) Show that the average weight, in grams, of the n loaves is given by

$$\frac{1}{2}(3n + 797)$$

There are 20 loaves in the display.

c) Using the standard results for summation, calculate the standard deviation, given by

$$\sqrt{\frac{\sum r^2}{n} - \left(\frac{\sum r}{n}\right)^2}$$

of the weights of the loaves in the display, giving your answer to 3 significant figures.

1.09 — Question 24.

a) Show that

$$\frac{1^3 + 2^3 + 3^3 + \ldots + n^3}{1 + 2 + 3 + \ldots + n} = \frac{1}{2}n(n + 1)$$

b) Hence, find the smallest value of n such that

$$\frac{1^3 + 2^3 + 3^3 + \ldots + n^3}{1 + 2 + 3 + \ldots + n} > 200$$

1.09 — Question 25.

a) Find a fully factorised expression for

$$\sum_{r=1}^{n}(r^3 - 2r)$$

b) Find the value of k such that

$$\sum_{r=1}^{k}(2r^3 + 3r) - \sum_{r=1}^{k}(r^3 + 5r) = 1224$$

Summation with Variable Lower Bounds

1.09 — Question 26. Rewrite each of the following in sigma notation starting at $r = 1$:

a) $\sum_{r=10}^{30}(r-15)(r-16)$

b) $\sum_{r=4}^{16}(r^2+3r)$

c) $\sum_{r=12}^{20}(3r^3-1)$

d) $\sum_{r=1}^{8}r^2 - \sum_{r=1}^{4}r^2$

1.09 — Question 27. Using the standard summation results, find:

a) $\sum_{r=10}^{20} (r^2 - 3)$

b) $\sum_{r=5}^{10} r(r-1)$

c) $\sum_{r=50}^{100} (r^3 - 1)$

1.09 — Question 28. Using the standard results for summations, evaluate $51^2 + 52^2 + 53^2 + \ldots + 101^2$

1.09 — Question 29.

a) Using the standard results for summations, show that

$$\sum_{r=1}^{n} r(r-1)(r+1) = \frac{1}{4}n(n-1)(n+1)(n+2)$$

b) Hence, show that

$$\sum_{r=1}^{2n} r(r-1)(r+1) = n(n+1)(2n-1)(2n+1)$$

1.09 — Question 30. Find fully factorised expressions for each of the following using standard series formulae:

a) $\sum_{r=1}^{2n} r^3$

b) $\sum_{r=n+1}^{2n} r^2$

c) $\sum_{r=1}^{n+4} r^2$

d) $\sum_{r=1}^{n+3} (r^3 - 2r^2)$

e) $\sum_{r=n+1}^{2n} (r^3 + r)$

1.09 — Question 31. Show that, for $n \geq 6$

$$\sum_{r=6}^{n+1} r^2 = \frac{1}{6}(an^3 + bn^2 + cn + d)$$

where a, b, c and d are constants to be found.

1.09 — Question 32.

a) Show that, for values of a and b to be found that

$$\sum_{r=1}^{n} r(r-5) = \frac{1}{3}n(n-a)(n+b)$$

b) Hence, find a fully factorised expression for

$$\sum_{r=n+1}^{3n} r(r-5)$$

1.09 — Question 33.

a) Find the sum of the squares of the first n positive even numbers.

b) Given that the sum of the squares of 5 consecutive even numbers is 1020, by using standard results for summation, determine the smallest of these consecutive even numbers.

1.09 — Question 34. Express

$$\sum_{r=1}^{2n} r^3 - \sum_{r=1}^{n} (r+2)^2$$

as a product of three factors in terms of n

1.09 — Question 35. Given a and b are constant integers, find the possible values of a and b such that

$$\sum_{r=1}^{n}(r+a)(r+b) = \frac{1}{3}n(n^2-7)$$

1.09 — Question 36.

a) Show that

$$\sum_{r=1}^{2n}(n-r)^2 = \frac{1}{3}n(2n^2+1)$$

b) Use standard summation results to prove that

$$\sum_{r=n+1}^{2n}(n-r)^2 = \sum_{r=1}^{n}r^2$$

1.09 — Question 37.

a) Show that for positive constants a, b, c and d

$$\sum_{r=1}^{3n-2}(5r+3) = \frac{1}{2}(an-b)(cn+d)$$

b) Hence calculate

$$\sum_{r=1}^{16}(5r+3)$$

1.09 — Question 38.

a) Show that

$$\sum_{r=1}^{2n+7}(r^2-1) = \frac{1}{3}(4n+19)(2n+7)(n+3)$$

b) Hence calculate

$$\sum_{r=10}^{63}(r^2-1)$$

Series with Alternating and Periodic Terms

1.09 — Question 39. Find:

a) $\sum_{r=1}^{8}(-1)^r$

b) $\sum_{r=1}^{2n}(-1)^r$

c) $\sum_{r=1}^{9}(-1)^r$

d) $\sum_{r=1}^{2n-1}(-1)^r$

1.09 — Question 40. Using the standard results for summations, show that:

a) $\sum_{r=1}^{2n}(-1)^r r = n$

b) $\sum_{r=1}^{2n-1}(-1)^r r = -n$

c) $\sum_{r=1}^{2n}(-1)^r(r+1) = n$

d) $\sum_{r=1}^{2n}(-1)^r r^2 = n(2n+1)$

Q41-Q44 use radians

1.09 — Question 41. Find, giving your answers in exact form:

a) $\displaystyle\sum_{r=1}^{10} \sin\left(\frac{\pi}{2}r\right)$

b) $\displaystyle\sum_{r=1}^{32} r\cos\left(\frac{\pi}{2}r\right)$

c) $\displaystyle\sum_{r=1}^{27} r\tan\left(\frac{\pi}{3}r\right)$

1.09 — Question 42. Find:

a) $\displaystyle\sum_{r=1}^{n} \sin(\pi r)$

b) $\displaystyle\sum_{r=1}^{n} \cos(2\pi r)$

c) $\displaystyle\sum_{r=1}^{3n} \tan\left(\frac{\pi}{3}r\right)$

1.09 — Question 43.

a) Show that

$$\sum_{r=1}^{n}(2r+1)^2 = \frac{1}{3}n(4n^2+12n+11)$$

b) Hence find the values of k for which

$$\sum_{r=1}^{k}(2r+1)^2 + 15\sum_{r=1}^{k}\cos(2\pi r) = 24k^2$$

1.09 — Question 44.

a) Show that

$$\sum_{r=1}^{n}(5r-3)^2 = \frac{1}{6}n(50n^2-15n-11)$$

b) Hence find the value of k for which

$$\sum_{r=3}^{k}(5r-3)^2 + 20\sum_{r=1}^{20} r\sin\left(\frac{\pi}{2}r\right) = 241k$$

1.09 — Question 45.

a) Using the standard results for summations, show that

$$\sum_{r=1}^{2n} r((-1)^r + 5)^2 = n(52n + 36)$$

b) Hence find

$$\sum_{r=15}^{80} r((-1)^r + 5)^2$$

1.09 — Question 46.

a) Using the standard results for summations, show that

$$\sum_{r=1}^{2n} ((-1)^r + 3r)^2 = 24n^3 + 18n^2 + 11n$$

b) Hence find

$$\sum_{r=49}^{100} ((-1)^r + 3r)^2$$

1.10 — Question 1. Find the sum of the following series. You do not have to simplify your answer.

a)
$$1-\frac{1}{2}+\frac{1}{2}-\frac{1}{3}+\frac{1}{3}-\frac{1}{4}+\cdots+\frac{1}{23}-\frac{1}{24}+\frac{1}{24}-\frac{1}{25}$$

b)
$$1-\frac{1}{4}+\frac{1}{4}-\frac{1}{7}+\frac{1}{7}-\frac{1}{10}+\cdots+\frac{1}{3n-5}-\frac{1}{3n-2}+\frac{1}{3n-2}-\frac{1}{3n+1}$$

c)
$$\frac{1}{6}-\frac{1}{10}+\frac{1}{8}-\frac{1}{12}+\frac{1}{10}-\frac{1}{14}+\frac{1}{12}-\frac{1}{16}\cdots$$
$$+\frac{1}{2(n+1)}-\frac{1}{2(n+3)}+\frac{1}{2(n+2)}-\frac{1}{2(n+4)}$$

d)
$$2-1+3-2+4-3+\cdots+n-(n-1)+(n+1)-n$$

e)
$$1\times 2-1\times 0+2\times 3-2\times 1+3\times 4-3\times 2+\cdots+n(n+1)-n(n-1)$$

1.10 — Question 2. Write each of the following as a single fraction:

a) $\frac{1}{n+1}-\frac{1}{n+2}$
b) $1+\frac{1}{n+1}$
c) $\frac{1}{2}-\frac{1}{n+2}$
d) $-\frac{3}{2}+\frac{1}{n+2}+\frac{1}{n+1}$
e) $\frac{1}{12}+\frac{1}{2(n+3)}-\frac{1}{2(n+2)}$

1.10 — Question 3. Use method of differences to find

$$\sum_{r=1}^{50}\left(\frac{1}{r+2}-\frac{1}{r}\right)$$

giving your answer in exact form.

1.10 — Question 4. Use method of differences to find

$$\sum_{r=8}^{100}\left(\frac{1}{r+1}-\frac{1}{r}\right)$$

giving your answer in exact form.

1.10 — Question 5. Use method of differences to find

$$\sum_{r=1}^{n}\left(\frac{1}{r+8}-\frac{1}{r+7}\right)$$

1.10 — Question 6. Use method of differences to show that

$$\sum_{r=1}^{n}\left(\frac{1}{r+3}-\frac{1}{r+4}\right)=\frac{n}{4(n+4)}$$

1.10 — Question 7. Use method of differences to find

$$\sum_{r=1}^{n}\left(\frac{1}{r+5}-\frac{1}{r+3}\right)$$

1.10 — Question 8.

a) Show that

$$\frac{1}{r+4}-\frac{1}{r+5}=\frac{1}{(r+4)(r+5)}$$

b) Hence find

$$\sum_{r=1}^{n}\frac{1}{(r+4)(r+5)}$$

1.10 — Question 9.

a) Write the following as a single fraction:

$$\frac{1}{r+1} - \frac{1}{r+2}$$

b) Hence show that

$$\sum_{r=1}^{n} \frac{1}{(r+1)(r+2)} = \frac{n}{an+b}$$

where a and b are constants to be found.

1.10 — Question 10.

a) Show that

$$\frac{1}{r-1} - \frac{1}{r+1} = \frac{2}{(r-1)(r+1)}$$

b) Hence find

$$\sum_{r=5}^{50} \frac{1}{r^2-1}$$

giving your answer in exact form.

1.10 — Question 11.

a) Show that

$$\frac{1}{r+3} - \frac{1}{r+4} = \frac{1}{r^2+7r+12}$$

b) Hence find

$$\sum_{r=8}^{22} \frac{1}{r^2+7r+12}$$

giving your answer in exact form.

1.10 — Question 12.

a) Show that

$$\frac{1}{r-1} - \frac{1}{r+2} = \frac{3}{(r-1)(r+2)}$$

b) Hence find

$$\sum_{r=2}^{n} \frac{1}{(r-1)(r+2)}$$

giving your answer in exact form.

1.10 — Question 13. By considering

$$\frac{1}{r^3} - \frac{1}{(r+1)^3}$$

find

$$\sum_{r=1}^{n} \frac{3r^2+3r+1}{r^3(r+1)^3}$$

1.10 — Question 14. By considering

$$\frac{1}{r-2} - \frac{1}{r}$$

find

$$\sum_{r=3}^{n} \frac{1}{r^2-2r}$$

as a single fraction.

1.10 — Question 15.

a) Find

$$\frac{1}{2r+1} - \frac{1}{2r+3}$$

as a single fraction.

b) Find the sum of the series

$$\frac{2}{3\times 5} + \frac{2}{5\times 7} + \frac{2}{7\times 9} + \cdots + \frac{2}{31\times 33}$$

giving your answer in exact form.

1.10 — Question 16.

a) Write

$$\frac{1}{r} - \frac{1}{r+1}$$

as a single fraction.

b) Find how many terms of the series

$$\frac{1}{1\times 2} + \frac{1}{2\times 3} + \frac{1}{3\times 4} + \cdots + \frac{1}{r(r+1)} + \cdots$$

are needed to exceed 0.98

1.10 — Question 17.

a) Write

$$\frac{1}{2r} - \frac{1}{2r+2}$$

as a single fraction.

b) Find

$$\sum_{r=1}^{n} \frac{1}{2r(r+1)}$$

c) Hence find

$$\sum_{r=1}^{\infty} \frac{1}{2r(r+1)}$$

1.10 — Question 18.

a) Show that

$$r+1 = \frac{1}{2}[(r+1)(r+2) - r(r+1)]$$

b) Hence use method of differences to find

$$\sum_{r=1}^{n} 2(r+1)$$

1.10 — Question 19.

a) Show that

$$(r+1)(r+2) \equiv \frac{1}{3}[(r+1)(r+2)(r+3) - r(r+1)(r+2)]$$

b) Hence use method of differences to find

$$\sum_{r=1}^{n} 3(r+1)(r+2)$$

1.10 — Question 20. Find the smallest integer n such that

$$\sum_{r=1}^{n} \left(\frac{2}{r^2} - \frac{2}{(r+1)^2} \right) > 1.95$$

1.10 — Question 21.

a) Show that

$$\frac{1}{r!} - \frac{1}{(r+1)!} = \frac{r}{(r+1)r!}$$

b) Hence show that

$$\sum_{r=1}^{n} \frac{r}{(r+1)r!} = a - \frac{b}{(n+1)!}$$

where a and b are constants to be found.

1.10 — Question 22.

a) Expand and simplify $(r+2)^3 - r^3$

b) Hence find

$$\sum_{r=1}^{n} 3r^2 + 6r + 4$$

giving your answer in fully factorised form.

c) By considering your answer to part b), find

$$\sum_{r=1}^{n} r(r+2)$$

1.10 — Question 23.

a) Show that

$$(2r+1)^3 - (2r-1)^3 = 24r^2 + 2$$

b) Hence, using method of differences, show that

$$\sum_{r=1}^{n} r^2 = \frac{1}{6}n(n+1)(2n+1)$$

1.10 — Question 24.

a) Show that

$$(2r+1)^4 - (2r-1)^4 = 64r^3 + 16r$$

b) Hence, using method of differences and the summation series formulae, show that

$$\sum_{r=1}^{n} r^3 = \frac{1}{4}n^2(n+1)^2$$

1.10 — Question 25.

a) Show that

$$(2r+1)^5 - (2r-1)^5 = 160r^4 + 80r^2 + 2$$

b) Hence, using method of differences and the summation series formulae, show that

$$\sum_{r=1}^{n} r^4 = \frac{1}{30}n(n+1)(2n+1)(3n^2+3n-1)$$

1.10 — Question 26. Find the value of k such that

$$\sum_{r=k}^{100}\left(\frac{2}{r} - \frac{2}{r+1}\right) = \frac{65}{1818}$$

1.10 — Question 27.

a) Show that

$$\frac{2}{r} - \frac{1}{r+1} - \frac{1}{r+2} = \frac{3r+4}{r(r+1)(r+2)}$$

b) Find

$$\sum_{r=1}^{n} \frac{3r+4}{r(r+1)(r+2)}$$

c) Find the value of k such that

$$\sum_{r=k}^{25} \frac{3r+4}{r(r+1)(r+2)} = \frac{1174}{12\,285}$$

1.10 — Question 28.

a) Show that

$$2^{r+1}(r-1) - 2^r(r-2) = r2^r$$

b) Hence show that

$$\sum_{r=1}^{n} r2^r = 2 + 2^{n+1}(n-1)$$

1.10 — Question 29.

a) Show that

$$\ln(r+2) - \ln(r+1) = \ln\left(1 + \frac{1}{r+1}\right)$$

b) Hence show that

$$\sum_{r=1}^{n} \ln\left(1 + \frac{1}{r+1}\right) = \ln\left(\frac{n+a}{b}\right)$$

where a and b are integers to be found.

1.10 — Question 30. Use method of differences to write

$$\sum_{r=1}^{n} \ln\left(\frac{r+1}{r}\right)$$

as a single function.

1.10 — Question 31. Find the exact value of

$$\sum_{r=21}^{50} \ln\left(\frac{r+1}{r-1}\right)$$

1.10 — Question 32.

a) Show that

$$\sum_{r=1}^{n} \ln\left(\frac{r+3}{r+1}\right) = \ln(a(n+b)(n+c))$$

where a, b and c are positive constants to be found.

b) Hence find

$$\sum_{r=1}^{n} \ln\left(\frac{r+3}{r+1}\right)^{36}$$

1.10 — Question 33.

a) Show that

$$\sum_{r=1}^{n} \ln\left(\frac{2r+3}{2r+1}\right) = \ln(a(bn+c))$$

where a, b and c are positive constants to be found.

b) Hence find the smallest value of n for which

$$\sum_{r=1}^{n} \ln\left(\frac{2r+3}{2r+1}\right) > 4$$

1.10 — Question 34. Find

$$\sum_{r=6}^{\infty} \left(\frac{1}{r+1} - \frac{1}{r+3}\right)$$

1.10 — Question 35.

a) Show that

$$\frac{1}{2(r+2)} - \frac{1}{2(r+4)} = \frac{1}{(r+2)(r+4)}$$

b) Show that if

$$\frac{7}{24} - \sum_{r=1}^{n} \frac{1}{(r+2)(r+4)} < \frac{19}{180}$$

then

$$n \geq 7$$

1.11 — Question 1. For each of the following, find the first 4 terms of the Maclaurin series and state the range of values for which the expansion is valid:

a) e^{3x}
b) e^{-2x}
c) $\ln(1+2x)$
d) $\ln(1-x)$
e) $\sin(2x)$
f) $\sin(-4x)$
g) $\cos\left(\frac{1}{2}x\right)$
h) $\cos(-x)$

1.11 — Question 2. Write down the coefficient of the second term of the Maclaurin series expansion for each of the following:

a) $\cos\left(\frac{5}{2}x\right)$

b) e^{-4x}

c) $\ln\left(1+\frac{1}{2}x\right)$

1.11 — Question 3.

a) Find the first three non-zero terms in the Maclaurin expansion of

$$\ln(3+x)$$

b) State the values for which the expansion is valid.

1.11 — Question 4.

a) Find the series expansion in ascending powers of x as far as the term in x^3 for

$$\ln\left(1+\frac{1}{4}x\right)$$

b) State the values of x for which the expansion is valid.

c) Hence find the series expansion up to and including the term in x^3 for

$$\ln\sqrt{1+\frac{1}{4}x}$$

1.11 — Question 5. Given that the coefficient of the third term in the Maclaurin series expansion of

$$\sin(px)$$

is $\frac{4}{15}$, find the value of p

1.11 — Question 6. Given that the coefficient of the third term in the Maclaurin series expansion of

$$e^{px}$$

is 72, find the possible values of p

1.11 — Question 7. Expand each of the following up to and including the x^3 term:

a) $\ln(4+4x)$
b) $\frac{3e^x+1}{e^x}$
c) $\ln(e-2x)$
d) $\frac{1}{x}\ln(1+x^2)$
e) $\frac{1}{x}e^{x^2}$
f) $\frac{1}{x}\ln(2-2x)$

1.11 — Question 8.

a) Find and simplify the first five non-zero terms of the Maclaurin Series for $\cos(2x)$

b) Hence write down the first five non-zero terms of the Maclaurin Series for $\cos(-2x)$

1.11 — Question 9.

a) Find and simplify the first four non-zero terms of the Maclaurin Series for $\sin(2x)$

b) Hence write down the first four non-zero terms of the Maclaurin Series for $\sin(-2x)$

1.11 — Question 10. Find the first three non-zero terms of the Maclaurin series for each of the following, in each case stating the values of x for which the expansion is valid:

a) $(4+4x)^{\frac{1}{2}}$

b) $(4+x)^{\frac{1}{2}}$

c) $\frac{1}{\sqrt{4+5x}}$

1.11 — Question 11. Find the series expansions, up to and including the term in x^3 of each of the following, in each case giving the range of values of x for which the expansion valid.

a) $\ln(3x^2 - x + 1)$

b) $\ln(2x^2 + 3x + 1)$

1.11 — Question 12.

a) Find the Maclaurin series expansion, up to and including the term in x^3, of

$$\ln\left(1+\frac{1}{2}x^2+4x\right)$$

b) Hence find the Maclaurin series expansion, up to and including the term in x^3, of

$$\ln(x^2+8x+2)$$

Sums of Maclaurin series

1.11 — Question 13. Find the first three non-zero terms of the Maclaurin series for each of the following:

a) e^x+e^{-x}
b) $\sin x+\cos x$
c) $\cos x-\sin x$
d) $e^x-\sin x$
e) $e^x-\cos x$
f) $2\sin 2x+2\cos 2x$
g) $e^{ax}+e^{-ax}$
where a is a positive constant.
h) $\sin ax+x\cos ax$
where a is a positive constant.

1.11 — Question 14. A function is given by

$$f(x) = e^{ax} + (1 + bx)^{\frac{1}{2}}$$

where a and b are positive constants.

a) State the range of values for which the expansion of $f(x)$ is valid in terms of b

b) Find the first four terms in the expansion of $f(x)$ in ascending powers of x, giving your answer in terms of a and b

It is given that the coefficient of x is -1 and the coefficient of x^2 is $\frac{5}{2}$

c) Find the values of a and b

d) Hence find the coefficient of x^3

Prerequisite knowledge for Q15-Q16: Hyperbolic Functions

1.11 — Question 15. By considering the first six non-zero terms of the Maclaurin series for e^{2x} find the first three terms of the Maclaurin series for $\sinh(2x)$

1.11 — Question 16. By considering the first five non-zero terms of the Maclaurin series for e^{2x} find the first three terms of the Maclaurin series for $\cosh(2x)$

Prerequisite knowledge for Q17-Q18: Hyperbolic Functions and Complex Numbers

1.11 — Question 17. By considering the first five non-zero terms of the Maclaurin series for e^{ix}

a) find the first three non-zero terms of the Maclaurin series for

$$\cosh(ix)$$

b) Hence, state a trigonometric function which has the same first three terms.

1.11 — Question 18. By considering the first six non-zero terms of the Maclaurin series for e^{ix}

a) find the first three terms of the Maclaurin series for

$$\sinh(ix)$$

It is known that $\sinh(ix) = i\mathrm{f}(x)$

b) State the function $\mathrm{f}(x)$

1.11 — Question 19. A function is given by

$$\mathrm{f}(x) = k(1+2x)^{\frac{1}{3}} - \ln(1+5x)$$

It is known that when $\mathrm{f}(x)$ is expanded as a series in ascending powers of x that the second term has coefficient -1. Find:

a) the value of k

b) the series expansion in ascending powers of x as far as the term in x^3

c) the values for which the expansion $\mathrm{f}(x)$ is valid.

1.11 — Question 20.

a) Fully factorise $1 - 2x - 8x^2$

b) Hence, find the first three non-zero terms of the Maclaurin series for

$$\ln(1 - 2x - 8x^2)$$

c) Determine the validity of

$$\ln(1 - 2x - 8x^2)$$

Products of Maclaurin series

1.11 — Question 21. Find the first three non-zero terms of the Maclaurin series for each of the following:

a) $e^x \sin x$

b) $e^{-x} \sin x$

c) $\sin x \ln(1 + x)$

d) $e^x \cos x$

e) $e^x \cos 2x$

1.11 — Question 22.

a) Find the first four terms in the expansion, in ascending powers of x of

$$(3+5x)^{-1}$$

b) State the values for which the expansion of $(3+5x)^{-1}$ is valid.

c) Hence, find the first four terms in the expansion, in ascending powers of x of

$$\mathrm{f}(x) = \frac{\sin 2x}{3+5x}$$

d) State the values for which the expansion of $\mathrm{f}(x)$ is valid.

1.11 — Question 23.

a) Find the Maclaurin expansion up to and including the x^5 term for both

$$\cos\theta \sin\theta \text{ and } \sin q\theta$$

where q is a positive constant.

b) Given that

$$\cos\theta \sin\theta = p \sin q\theta$$

where p and q are positive constants, find the value of p and q

1.11 — Question 24. Given that the coefficient of the third term in the Maclaurin series expansion of

$$e^{px}\sin x$$

is $\frac{13}{3}$
Find the possible values of p

1.11 — Question 25. Given that the coefficient of the third term in the Maclaurin series expansion of

$$e^{x}\sin px$$

is $-\frac{1}{3}$
Find the possible values of p

1.11 — Question 26. Given that the coefficient of the third term in the Maclaurin series expansion of

$$\frac{\cos px}{e^x}$$

is -12
Find the possible values of p

1.11 — Question 27. Find the values of a, b and k given that

$$\frac{\sin ax}{be^x} = \frac{1}{2}x - kx^2 - \frac{1}{12}x^3 + \cdots$$

where a, b and k are positive constants.

Using Maclaurin series

1.11 — Question 28. Use the appropriate Maclaurin series to find the answer correct to 4 decimal places, for each of the following:

a) $e^{0.1}$
b) $\ln(0.9)$
c) $\sin\left(\frac{\pi}{18}\right)$
d) $\cos(14^\circ)$

1.11 — Question 29. Samara is using a Maclaurin series to find a numerical value for $\ln 2.7$
Her workings are below:

$$\begin{aligned}\ln(1+1.7) &\approx 1.7 - \frac{1.7^2}{2} + \frac{1.7^3}{3} - \frac{1.7^4}{4} + \cdots \\ &\approx 1.7 - 1.445 + 1.6376\ldots - 2.088025 + \cdots \\ &= -0.1953 \text{ to 4 decimal places.}\end{aligned}$$

a) Use your calculator to find $\ln 2.7$
b) Explain why Samara's method isn't correct.
c) Given $\ln 3 \approx 1.09861$, use a Maclaurin series to find $\ln 2.7$ to 4 decimal places.

1.11 — Question 30.

a) Find the series expansion of $\frac{1}{\sqrt{e^x}}$ up to and including the term in x^5

b) Hence find the value of $\frac{1}{\sqrt{e}}$ to 4 decimals places.

1.11 — Question 31. By using a Maclaurin series up to the third non-zero term, find each of the following to two decimal places:

a) $\int_0^{0.5} e^x \, dx$

b) $\int_0^{0.5} \ln(1+x) \, dx$

c) $\int_0^{1.5} \sin x \, dx$

d) $\int_0^{1.5} \cos x \, dx$

e) $\int_0^{0.5} (1-x)^{\frac{1}{2}} \, dx$

1.11 — Question 32. By using a Maclaurin series up to the x^3 term, find an approximation to

$$\int \frac{1}{\sqrt{1-x}}\,dx$$

1.11 — Question 33.

a) Find a cubic approximation for

$$\ln(1+2x)$$

b) Hence use your approximation to find a non-zero approximate solution to

$$\ln(1+2x) = 8x^2$$

Give your answer to 3 significant figures.

c) Suggest an improvement to your approximation in part b) and justify your answer.

1.11 — Question 34.

a) Find a quartic approximation for $\cos\theta$

b) Hence use your approximation to find an approximate positive acute solution to

$$\cos\theta = -\frac{1}{648}\theta^4$$

Give your answer to 3 significant figures.

1.11 — Question 35. It is known that

$$\mathrm{f}(x) = e^x - \cos x$$

has a root near $x = 0$

a) Find the Maclaurin series expansion of $\mathrm{f}(x)$ up to and including the term in x^3

b) Hence find an estimate for the non-zero root of $\mathrm{f}(x) = 0$, that is the closest to 0, giving your answer to 3 significant figures.

1.11 — Question 36. Using $x = 0.2$ in the expansion of

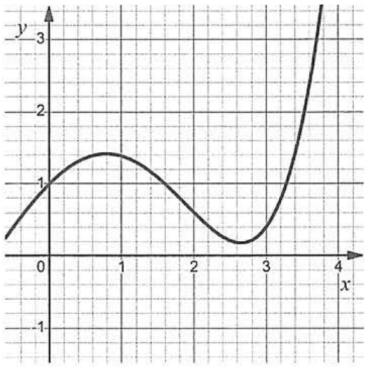
$$\ln\left(\frac{1+x}{1-x}\right)$$

a) find $\ln 1.5$ correct to 3 decimals places.
b) Find the percentage error of your solution to part a) with $\ln 1.5$ on your calculator.

1.11 — Question 37.
a) Find the first four terms in the expansion, in ascending powers of x of $e^x\sqrt{1+2x}$
b) Hence use $x = 0.5$ to find an approximation to $\sqrt{2}e^{\frac{1}{2}}$ giving your answer to 3 significant figures.
c) Find the percentage error between your answer in part b) and the exact value.

1.11 — Question 38. By considering the Maclaurin expansion of both e^{-x} and $\cos(x)$ up to and including the x^2 term, sketch the graph of $y = e^{-x} - \cos x$ for $0 \leq x \leq 0.4$

1.11 — Question 39. Two distinct functions $\mathrm{f}(x)$ and $\mathrm{g}(x)$ have been reduced to the first three terms of their Maclaurin series and $\mathrm{f}(x) + \mathrm{g}(x)$ is shown in the graph below. The functions are two of e^x, $\cos x$, $\sin x$ and $\ln(1+x)$



Given that by including the fourth term of both the Maclaurin series for the curve is negative for values larger than $x \approx 2.3376$, identify the functions $\mathrm{f}(x)$ and $\mathrm{g}(x)$

1.11 — Question 40. A triangle ABC has sides of lengths $AB = 2$ and $AC = 3$, and the angle BAC is $2x$. Show that the area of ABC is approximately $px - qx^3$ where p and q are positive constants.
State the values of p and q

1.11 — Question 41.

a) Find the series expansion in ascending powers of x as far as the term in x^3 for

$$\frac{\cos x}{\sqrt{1+x}}$$

b) Use your approximation in part a) to find an approximation of the equation of the tangent to the curve at $x = 0$ of $y = \frac{\cos x}{\sqrt{1+x}}$

1.11 — Question 42.

a) Find the Maclaurin series of $\sin^2(2x)$ up to and including the term in x^4

b) Hence use your approximation to find a non-zero approximate solution to

$$\sin^2(2x) = 16x^3$$

Give your answer to 3 significant figures.